



Selection of transport aircraft using the hybrid CRITIC-TOPSIS method and imputation of missing values in a neutrosophic environment

Gurpreet Kaur¹ · Arunava Majumder² · Aishwarya Dhara³ · Laxmipriya Murmu⁴ · Balvinder Singh Sandhu¹ · Manoj Singh Adhikari⁵ · P. Uma Maheswari²

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Abstract

The selection of optimal transport aircraft is a complex process that involves several conflicting criteria and uncertainty. The transport aircraft industry has been booming in recent days, and the selection of suitable aircraft is one of the foremost concerns to maintain higher profitability in the aviation sector. In this regard, this paper develops a CRITIC-TOPSIS “multi-criteria decision-making” method with an incomplete dataset (consisting of missing values).

The objective of the paper is to impute the missing entries with suitable neutrosophic values and to extend the decision-making process with hybrid CRITIC-TOPSIS under a neutrosophic environment. The study utilizes empirical data for 38 different transport aircraft as alternatives, evaluating them against 10 critical criteria. To impute the null values in the existing dataset, the neutrosophic set is used, and the hybrid CRITIC-TOPSIS method is utilized for decision-making purposes.

Keywords Transport aircraft selection · CRITIC-TOPSIS · Neutrosophic sets · Multi-criteria decision-making · Uncertainty modeling

✉ Arunava Majumder
am.arunavamajumder@gmail.com

✉ Aishwarya Dhara
dharaaishwarya@gmail.com

Gurpreet Kaur
gurpreet.kaur@lpu.co.in

Laxmipriya Murmu
laxmipriyam1405@gmail.com

Balvinder Singh Sandhu
balvinder.singh@lpu.co.in

Manoj Singh Adhikari
manoj.space99@gmail.com

P. Uma Maheswari
umamahep@srmist.edu.in

¹ Department of Mathematics, Lovely Professional University, Phagwara, Punjab 144411, India

² Department of Mathematics, College of Engineering and Technology, SRM Institute of Science and Technology, Kattankulathur, Chennai, Tamil Nadu 603203, India

³ Department of Aerospace Engineering, School of Aeronautical Sciences, Hindustan Institute of Technology and Science, Padur, Chennai 603103, India

⁴ Department of Aerospace Engineering, Lovely Professional University, Phagwara, Punjab 144411, India

⁵ Department of Computer Science & Engineering, Graphic Era Hill University, Haldwani, Uttarakhand 263139, India

1 Introduction

Transport aircraft selection is a highly complex and difficult procedure that requires many factors to be considered. The multiple criteria that affect the selection of the best aircraft include range, fuel efficiency, payload capacity, operational reliability, etc. To enhance the efficiency of the selection process, researchers now highly rely on multi-criteria decision-making (MCDM) processes. MCDM helps choose the most suitable alternative when multiple constraints or criteria exist (Kaur et al. 2022, 2023). Recent studies have shown the utility of combined quantitative performance metrics and qualitative expert evaluations for the best aircraft selection with multidimensional nature (Khan et al. 2023; Li and Zhang 2022). Moreover, uncertainty and data vagueness are significant issues that can lead to inappropriate decision-making. This problem can be well encountered by integrating sustainability factors and uncertainty modeling—particularly through fuzzy, intuitionistic, or neutrosophic approaches (Bose et al. 2021; Yadav et al. 2023). In this regard, the applicability of hybrid decision-making frameworks like CRITIC-TOPSIS is highly encouraged that

merges the identification of criteria weights and ranking process to facilitate a more resilient aircraft selection method (Singh and Kumar 2022). The neutrosophic approach is also useful for uncertainty conditions, such as missing data inputs. This technique helps identify the missing values more efficiently than the traditional statistical approach.

Uncertainty modelling is a critical part in strategic sectors, like the selection of suitable aircraft under various conditions or criteria provided. The traditional multi-criteria

decision-making (MCDM) approach may be insufficient in making strategic decisions under uncertain conditions that involve numerous quantitative and qualitative data. Over the past ten years, a variety of studies have been conducted to integrate the ambiguity and vagueness of the initial data into multi-criteria decision-making (MCDM) techniques for the resolution of complicated real-world issues (Boran et al. 2009; Kaya and Kahraman 2011). Traditional decision-making methods under uncertainty, commonly available in the existing literature are fuzzy sets, intuitionistic fuzzy sets, and more recently, neutrosophic sets (Atanassov 1986; Wang and Li 2010; Zadeh 1965). Smarandache (Smarandache 2005) introduced the Neutrosophic sets as an extension of fuzzy and intuitionistic fuzzy sets that explicitly model truth, indeterminacy, and falsity independently. This structure can efficiently handle uncertainty more appropriately in the presence of critical datasets such as performance matrices or missing values determination (Ye 2014; Li and Liu 2014).

One of the most significant problems in real-world data analysis and decision-making is missing values in the datasets, which has a huge impact on obtaining predictions and the right decision. However, there are numerous missing value imputation processes, such as mean, median, and mode imputation, that are purely central tendency and may not depict the original scenario as well. To tackle this limitation, the concept of fuzzy, intuitionistic, and neutrosophic sets can be a powerful tool to handle the uncertainty. Missing value finding in an inappropriate dataset is one of the most important areas of research. The use of statistical methods and fuzzy approaches was discussed in various literature (Smarandache 2014). Recently, a combined approach of neutrosophic sets and machine learning algorithms claims to handle the uncertainty in incomplete data efficiently, obtaining improved results in decision-making and prediction modelling (Kamacı 2021). Moreover, innovative methodologies for risk ranking utilizing single-valued neutrosophic sets have incorporated data imputation strategies to address absent information throughout the aggregation procedure, thereby underscoring the efficacy of neutrosophic frameworks in facilitating decision-making amidst uncertainty (Abdel-Basset et al. 2021). A tabulated illustration of contributions of the previous literature is given in Table 1.

1.1 Essential definitions

1.1.1 Neutrosophic set

Let U be a universal set (non-empty and possibly infinite). A neutrosophic set $\mathcal{N}$ in U is expressed as

Table 1 Author's contribution table of literature review

Article	Contribution	Research gap with this paper
Kaur et al. (2020)	Proposed intuitionistic fuzzy approach for energy power plant location selection	Limited to intuitionistic fuzzy sets, doesn't handle indeterminacy as comprehensively as neutrosophic sets
Kaur et al. (2023)	Developed CRITIC-TOPSIS under neutrosophic environment for aircraft selection	Focused only on complete datasets without addressing missing value imputation
Kaur et al. (2022)	Generalized fuzzy TOPSIS for hybrid energy resource selection	Used traditional fuzzy sets without considering truth-indeterminacy-falsity components
Khan et al. (2023)	Integrated MCDM framework for transport aircraft selection under uncertainty	Lacked specific focus on neutrosophic sets and missing data handling
Li and Zhang (2022)	Sustainable aircraft selection using life cycle cost and performance criteria	Did not incorporate uncertainty modeling through advanced fuzzy sets
Bose et al. (2021)	Intuitionistic fuzzy AHP-TOPSIS for aircraft selection	Used intuitionistic fuzzy sets which are less expressive than neutrosophic sets
Yadav et al. (2023)	Decision support system using neutrosophic sets and hybrid MCDM	Did not combine CRITIC weight determination with TOPSIS ranking
Singh and Kumar (2022)	Hybrid CRITIC-TOPSIS under fuzzy environment	Used conventional fuzzy sets rather than neutrosophic framework
Boran et al. (2009)	Intuitionistic fuzzy group decision making for supplier selection	Limited to intuitionistic fuzzy approach without neutrosophic extension
Kaya and Kahraman (2011)	Modified fuzzy TOPSIS for energy planning	Did not address missing data or use objective weighting methods like CRITIC
Smarandache (2005)	Introduced neutrosophic sets theory	Theoretical foundation without specific application to aircraft selection
Abdel-Basset et al. (2021)	Risk ranking with single-valued neutrosophic sets	Focused on risk assessment rather than comprehensive MCDM for selection
Diakoulaki et al. (1995)	Developed original CRITIC method for objective weighting	Used crisp numbers without fuzzy or neutrosophic extensions

$$\mathcal{N} = \{\langle u, \mu_{\mathcal{N}}(u), \iota_{\mathcal{N}}(u), \nu_{\mathcal{N}}(u) \rangle \mid u \in U\} \quad (1)$$

where

$\mu_{\mathcal{N}}$: truth-membership function,
 $\iota_{\mathcal{N}}$: indeterminacy function,
 $\nu_{\mathcal{N}}$: falsity/non-membership function, and

$$\mu_{\mathcal{N}}, \iota_{\mathcal{N}}, \nu_{\mathcal{N}} : U \rightarrow]0^-, 1^+[$$

with values taken from a non-standard interval $]0^-, 1^+[$. Furthermore,

$$0^- \leq \sup \mu_{\mathcal{N}}(u) + \sup \iota_{\mathcal{N}}(u) + \sup \nu_{\mathcal{N}}(u) \leq 3^+ \quad (2)$$

1.1.2 Single-valued neutrosophic set (SVNS)

Given a universe U , a SVNS $\mathcal{S}$ over U is defined as

$$\mathcal{S} = \{\langle u, \tau_{\mathcal{S}}(u), \epsilon_{\mathcal{S}}(u), \phi_{\mathcal{S}}(u) \rangle \mid u \in U\} \quad (3)$$

with $\tau_{\mathcal{S}}$: truth-membership, $\epsilon_{\mathcal{S}}$: indeterminacy, $\phi_{\mathcal{S}}$: non-membership, and

$$\tau_{\mathcal{S}}, \epsilon_{\mathcal{S}}, \phi_{\mathcal{S}} : U \rightarrow [0, 1]$$

satisfying

$$0 \leq \sup \tau_{\mathcal{S}}(u) + \sup \epsilon_{\mathcal{S}}(u) + \sup \phi_{\mathcal{S}}(u) \leq 3 \quad (4)$$

If $U = \{u\}$, i.e., a singleton set, then the SVNS is known as a *single-valued neutrosophic number (SVNN)*.

In simplified form:

$$\mathcal{S} = (u, \tau, \epsilon, \phi)$$

1.2 Set Operations on SVNSs

Let $\mathcal{A}, \mathcal{B}$ be SVNSs over universe U . Then,

- $\mathcal{A} = \mathcal{B} \iff \tau_{\mathcal{A}}(u) = \tau_{\mathcal{B}}(u), \epsilon_{\mathcal{A}}(u) = \epsilon_{\mathcal{B}}(u), \phi_{\mathcal{A}}(u) = \phi_{\mathcal{B}}(u)$
- $\mathcal{A} \subseteq \mathcal{B} \iff \tau_{\mathcal{A}}(u) \leq \tau_{\mathcal{B}}(u), \epsilon_{\mathcal{A}}(u) \leq \epsilon_{\mathcal{B}}(u), \phi_{\mathcal{A}}(u) \geq \phi_{\mathcal{B}}(u)$
- $\mathcal{A}^c = \{\langle u, \phi_{\mathcal{A}}(u), 1 - \epsilon_{\mathcal{A}}(u), \tau_{\mathcal{A}}(u) \rangle \mid u \in U\}$
- $\mathcal{A} \setminus \mathcal{B} = \{\langle u, \min(\tau_{\mathcal{A}}, \phi_{\mathcal{B}}), \min(\epsilon_{\mathcal{A}}, 1 - \epsilon_{\mathcal{B}}), \max(\phi_{\mathcal{A}}, \phi_{\mathcal{B}}) \rangle \mid u \in U\}$
- $\mathcal{A} \cap \mathcal{B} = \{\langle u, \min(\tau_{\mathcal{A}}, \tau_{\mathcal{B}}), \min(\epsilon_{\mathcal{A}}, \epsilon_{\mathcal{B}}), \min(\phi_{\mathcal{A}}, \phi_{\mathcal{B}}) \rangle \mid u \in U\}$
- $\mathcal{A} \cup \mathcal{B} = \{\langle u, \max(\tau_{\mathcal{A}}, \tau_{\mathcal{B}}), \max(\epsilon_{\mathcal{A}}, \epsilon_{\mathcal{B}}), \max(\phi_{\mathcal{A}}, \phi_{\mathcal{B}}) \rangle \mid u \in U\}$

Further properties:

- $\mathcal{A} \subseteq \mathcal{B}$ and $\mathcal{B} \subseteq \mathcal{C} \Rightarrow \mathcal{A} \subseteq \mathcal{C}$
- $\mathcal{A} \subseteq \mathcal{B} \Rightarrow \mathcal{A}^c \subseteq \mathcal{B}^c$
- $\mathcal{A}, \mathcal{C} \subseteq \mathcal{B} \Rightarrow \mathcal{A} \cup \mathcal{C} \subseteq \mathcal{B}$
- $\mathcal{A} \subseteq \mathcal{B}$ and $\mathcal{A} \subseteq \mathcal{C} \Rightarrow \mathcal{A} \subseteq \mathcal{B} \cap \mathcal{C}$
- $\mathcal{A} \subseteq \mathcal{B}, \mathcal{C} \subseteq \mathcal{D} \Rightarrow \mathcal{A} \cap \mathcal{C} \subseteq \mathcal{B} \cap \mathcal{D}$

Definition 1 $\mathcal{S}$ is called a universal SVNS over U if for all $u \in U$,

$$\tau_{\mathcal{S}}(u) = \epsilon_{\mathcal{S}}(u) = 1, \quad \phi_{\mathcal{S}}(u) = 0$$

Definition 2 Given two SVNNs: $\mathcal{S}_1 = (\tau_1, \epsilon_1, \phi_1)$ and $\mathcal{S}_2 = (\tau_2, \epsilon_2, \phi_2)$, their sum is defined as

$$\mathcal{S}_1 \oplus \mathcal{S}_2 = (\tau_1 + \tau_2 - \tau_1 \tau_2, \epsilon_1 \epsilon_2, \phi_1 \phi_2)$$

Definition 3 The product of two SVNNs $\mathcal{S}_1$ and $\mathcal{S}_2$ is given by:

$$\mathcal{S}_1 \otimes \mathcal{S}_2 = (\tau_1 \tau_2, \epsilon_1 + \epsilon_2 - \epsilon_1 \epsilon_2, \phi_1 + \phi_2 - \phi_1 \phi_2)$$

Definition 4 For any scalar $t > 0$, and a SVNN $\mathcal{S} = (\tau, \epsilon, \phi)$,

$$t\mathcal{S} = (1 - (1 - \tau)^t, \epsilon^t, \phi^t)$$

Definition 5 Let $X = (X_1, \dots, X_n), Y = (Y_1, \dots, Y_n)$ be two SVNS vectors over U . The neutrosophic distance is

$$D(X, Y) = \sqrt{\frac{1}{3n} \sum_{i=1}^n [(\tau_{X_i} - \tau_{Y_i})^2 + (\epsilon_{X_i} - \epsilon_{Y_i})^2 + (\phi_{X_i} - \phi_{Y_i})^2]}$$

Definition 6 Score function of a SVNN $\mathcal{S} = (\tau, \epsilon, \phi)$ is:

$$\delta(\mathcal{S}) = \frac{\tau - 2\epsilon - \phi + 3}{4}$$

Definition 7 Correlation coefficient for SVNSs Broumi and Smarandache (2013) Given two neutrosophic sets $\mathcal{N}_1$ and $\mathcal{N}_2$ over $X = \{x_1, \dots, x_n\}$, define:

$$\rho(\mathcal{N}_1, \mathcal{N}_2) = \frac{C(\mathcal{N}_1, \mathcal{N}_2)}{\sqrt{V(\mathcal{N}_1) \cdot V(\mathcal{N}_2)}}$$

where

$$C(\mathcal{N}_1, \mathcal{N}_2) = \sum_{i=1}^n [\tau_1(x_i)\tau_2(x_i) + \epsilon_1(x_i)\epsilon_2(x_i) + \phi_1(x_i)\phi_2(x_i)]$$

$$V(\mathcal{N}_j) = \sum_{i=1}^n [\tau_j^2(x_i) + \epsilon_j^2(x_i) + \phi_j^2(x_i)] \quad \text{for } j = 1, 2$$

1.3 Additional results

1. Symmetry: $C(\mathcal{N}_1, \mathcal{N}_2) = C(\mathcal{N}_2, \mathcal{N}_1)$, hence

$$\rho(\mathcal{N}_1, \mathcal{N}_2) = \rho(\mathcal{N}_2, \mathcal{N}_1)$$

2. Normalization:

$$0 \leq \rho(\mathcal{N}_1, \mathcal{N}_2) \leq 1$$

2 Problem statement

This paper proposes a hybrid approach for handling uncertainty in missing value imputation and selecting the best alternative using the CRITIC-TOPSIS method within a neutrosophic environment. A single-valued neutrosophic set (SVNS) performs the missing value imputation. CRITIC is applied to obtain the objective weights of the criteria supplied by the dataset. TOPSIS is used to get the rank of the alternatives based on their distance to the ideal and the anti-ideal solutions (Hwang and Yoon 1981). This cohesive methodology guarantees a thorough examination of uncertainty and augments the dependability and resilience of decision-making results in critical transportation scenarios (Liu and Zhang 2021; Peng and Wang 2018).

In this study, we considered 33 types of transport aircraft models of different companies: AIRBUS, BOEING, McDonnell Douglas, British Aerospace, Embraer, and Fokker. All 38 aircraft models are considered as the alternatives and 10 features related to the aircraft performance are assumed as the criteria. The 10 features are Static thrust (kN), Max. take-off (MTOW), Finess Ratio, Aspect Ratio,

Max. wing load (kg/m²), Thrust/Weight Ratio, Design range, Take-off distance (S_{TO}), Speed (kt), and Fuel consumption (kg/h). The missing values are available on the Design range, Take-off distance (S_{TO}), Speed (kt), and Fuel consumption (kg/h) features. We aim to deal with the uncertainty for each parameter with neutrosophic values. The missing values are also replaced with the neutrosophic values with suitable methods (discussed in the methodology section).

3 Methodology

The methodology of this study is divided into several parts as a hybrid method is used to create an efficient 'decision support system'. The methodology framework is distributed among the following points.

3.1 Determining criteria weights via the neutrosophic CRITIC method

To incorporate inconsistency and indeterminacy in decision-making, Smarandache (2005) introduced the concept of neutrosophic sets (NS). In multi-criteria decision-making (MCDM) problems, the ranking of alternatives is fundamentally dependent on the relevance of the criteria selected. Therefore, assigning appropriate weights to each criterion is critical.

One of the prominent objective weighting techniques is the CRITIC (CRiteria Importance Through Intercriteria Correlation) method, introduced by Diakoulaki et al. (1995). This method assesses the discriminating power of each criterion by utilizing standard deviation and correlation analysis. A higher standard deviation indicates a greater ability to differentiate among alternatives, while the correlation coefficient quantifies the degree of conflict between criteria (Rani et al. 2021; Yilmaz and Harmancioglu 2010).

In this study, the classical CRITIC method is extended into a neutrosophic context (N-CRITIC), incorporating truth-membership, indeterminacy-membership, and falsity-membership functions. The complete procedure is illustrated in Figure 1.

3.2 Step 1: Formulation of the Initial Decision Matrix

Let $\mathcal{A} = \{A_1, A_2, \dots, A_n\}$ represent the set of alternatives and $\mathcal{C} = \{C_1, C_2, \dots, C_m\}$ the set of criteria. The initial decision matrix, composed of crisp evaluation scores x_{ij} for each alternative A_i under criterion C_j , is shown in Table 2.

CRITIC METHOD

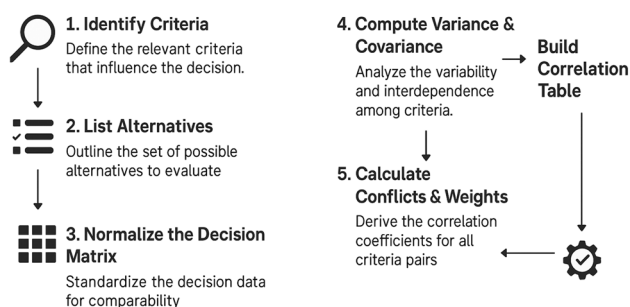


Fig. 1 Steps involved in the Neutrosophic CRITIC Method

Table 2 Initial Decision Matrix (Crisp Values)

Alternatives	C_1	C_2	...	C_m
A_1	x_{11}	x_{12}	...	x_{1m}
A_2	x_{21}	x_{22}	...	x_{2m}
			$\ddots$	
A_n	x_{n1}	x_{n2}	...	x_{nm}

3.3 Step 2: Construction of the Normalized Decision Matrix

The decision matrix is normalized to make the criteria comparable. We segregate the criteria as favorable and unfavorable. In favorable criteria, a larger value indicates better performance, and in unfavorable criteria, a smaller value indicates the same. The normalized value $\tilde{x}_{ij}$ is computed as:

$$\tilde{x}_{ij} = \frac{x_{ij} - x_j^{worst}}{x_j^{best} - x_j^{worst}}, \quad 1 \leq i \leq n, 1 \leq j \leq m \quad (5)$$

Here, x_j^{worst} denotes the highest value in the unfavorable column and the lowest value of the favorable column. Whereas, x_j^{best} denotes opposite.

3.4 Step 3: Neutrosophic Representation of the Normalized Matrix

Each normalized value $\tilde{x}_{ij}$ is transformed into a single-valued neutrosophic number (SVNN) defined as:

$$\tilde{x}_{ij} \longrightarrow \langle T_{ij}, I_{ij}, F_{ij} \rangle$$

where $T_{ij}, I_{ij}, F_{ij} \in [0, 1]$ denote the truth, indeterminacy, and falsity memberships, respectively. The following method is used to obtain the neutrosophic values and missing value imputation.

For missing value, we use truth, indeterminacy, and falsity as 0,1,1, i.e., the maximum indeterminacy. For, other cases, indeterminacy is taken as lower values as 0.1 to 0.3 for modern models, and higher for older models (0.4 to 0.6).

3.5 Step 4: Computation of Correlation Coefficients

As conventional correlation cannot be directly applied to neutrosophic numbers, pairwise correlation analysis is conducted among all criteria based on neutrosophic similarity or adapted correlation measures (Karaaslan 2019).

3.6 Step 5: Derivation of Criteria Weights

For each criterion C_j , the conflict intensity and informational content are determined by:

$$\gamma_j = \sigma_j \sum_{k=1}^m (1 - r_{jk}) \quad (6)$$

where σ_j is the standard deviation of criterion C_j and r_{jk} is the correlation coefficient between criteria C_j and C_k . Finally, the normalized objective weight w_j is calculated as:

$$w_j = \frac{\gamma_j}{\sum_{j=1}^m \gamma_j}, \quad j = 1, 2, \dots, m \quad (7)$$

4 Neutrosophic TOPSIS methodology

The neutrosophic TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) method ranks alternatives relative to the ideal solution. It is grounded in identifying the alternative closest to the neutrosophic positive ideal solution (NPIS) and farthest from the neutrosophic negative ideal solution (NNIS).

4.1 Step 1: normalized neutrosophic decision matrix

This matrix is reused from the N-CRITIC method.

4.2 Step 2: weighted normalized neutrosophic matrix

Each neutrosophic value $\langle T_{ij}, I_{ij}, F_{ij} \rangle$ is multiplied by its corresponding weight w_j as:

$$\tilde{x}_{ij}^{(w)} = \langle 1 - (1 - T_{ij})^{w_j}, I_{ij}^{w_j}, F_{ij}^{w_j} \rangle \quad (8)$$

Table 3 Table for selection criteria

Model	Static Thrust (kN) (F)	MT-OW (UF)	Fine-ness Ratio (F)	Asp-ect Ratio (F)	Wing Load (Kg/m ²) (UF)	Thr-ust/Wei-ght Ratio (F)	Range (km) (F)	Take-off Dist. (m) (UF)	Speed (kt) (F)	Fuel Cons. (kg/h) (UF)
A300-600R	257.0	170500	9.45	7.73	655.77	0.3073	4000	3189	480	5120
A310-300	231.0	150000	8.00	8.80	684.93	0.3140	4300	2450	484	4690
A319-100	99.7	64000	8.57	9.39	522.88	0.3176	1900	2080	487	3160
A320-200	111.2	73500	9.51	9.39	600.49	0.3084	2700	2590	487	3200
A340-500	235.8	365000	11.63	8.56	834.67	0.2634	8500	3550	—	—
A340-600	249.1	365000	12.34	8.56	834.67	0.2783	7500	3550	—	—
707-320C	84.5	151315	7.30	6.96	533.93	0.2278	—	—	521	—
727-200Adv	71.2	95028	7.00	6.86	601.82	0.2291	2400	3658	530	4536
737-400	97.9	62820	7.40	9.17	690.03	0.3177	2700	2475	492	3307
737-500	82.3	52390	7.40	9.17	575.46	0.3203	1700	2003	492	3574
737-700	89.0	69400	7.40	9.44	556.98	0.2615	3197	2042	—	—
737-800	107.0	78220	7.40	9.44	627.77	0.2789	2897	2316	—	—
767-200ER	257.6	175540	9.39	7.99	619.63	0.2992	6805	3080	488	5015
767-300ER	273.6	181437	10.67	7.99	640.44	0.3074	5760	3996	486	5171
DC8-73	98.0	161000	14.66	7.52	592.13	0.2480	—	3139	479	4881
DC 9-10	54.5	35245	7.75	8.56	406.19	0.3150	1311	—	501	—
DC 9-30	62.0	49000	8.97	8.72	527.05	0.2580	1725	2000	503	4050
DC 9-40	68.9	51710	9.64	8.72	556.20	0.2720	1685	2402	501	3980
DC 9-50	68.9	54420	10.32	8.72	585.35	0.2580	1500	3130	501	4525
DC10-10	178.0	190854	8.63	6.91	519.05	0.2850	4000	3816	475	9445
DC10-30	236.0	263636	8.63	6.91	716.99	0.2740	5373	3194	530	9743
DOUG-MD-81	85.6	63500	11.44	9.62	565.45	0.2750	1564	2646	499	4077
DOUG-MD-82	92.7	67812	11.44	9.62	603.85	0.2790	2050	2612	499	4077
DOUG-MD-83	96.5	72580	11.44	9.62	646.30	0.2710	2502	3066	499	4027
DOUG-MD-87	92.7	63500	10.06	9.62	565.45	0.2980	2372	2161	499	3924
DOUG-MD-90-30	111.2	70760	11.91	9.62	630.10	0.3200	2275	—	—	—
DOUG-MD-11	274.0	283720	9.74	7.91	837.18	0.2950	6787	3078	580	8970
RJ70	27.3	40823	6.74	8.89	528.11	0.2724	1952	1791	432	2150
RJ85	31.1	42184	7.44	8.89	545.72	0.3010	1782	1996	432	2237
RJ100	31.1	44225	8.12	8.89	572.12	0.2871	1671	2118	432	2310
RJ115	31.1	46040	8.12	8.89	595.60	0.2758	1794	2118	432	2380
F70	61.6	36740	8.45	8.43	392.94	0.3418	1080	1434	461	2391
F100	61.6	43090	9.85	8.43	460.86	0.2915	1290	2307	456	2565

F: Favorable; UF: Unfavorable

4.3 Step 3: identification of NPIS and NNIS

Let $\mathcal{C}_B$ denote the set of benefit-type criteria and $\mathcal{C}_C$ the set of cost-type criteria. The NPIS $V^+ = \langle T_j^+, I_j^+, F_j^+ \rangle$ and NNIS $V^- = \langle T_j^-, I_j^-, F_j^- \rangle$ are calculated as:

$$\begin{aligned}
 T_j^+ &= \begin{cases} \max_i T_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_B \\ \min_i T_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_C \end{cases} & T_j^- &= \begin{cases} \min_i T_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_B \\ \max_i T_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_C \end{cases} \\
 I_j^+ &= \begin{cases} \min_i I_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_B \\ \max_i I_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_C \end{cases} & I_j^- &= \begin{cases} \max_i I_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_B \\ \min_i I_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_C \end{cases} \\
 F_j^+ &= \begin{cases} \min_i F_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_B \\ \max_i F_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_C \end{cases} & F_j^- &= \begin{cases} \max_i F_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_B \\ \min_i F_{ij}^{(w)} & \text{if } C_j \in \mathcal{C}_C \end{cases}
 \end{aligned}$$

4.4 Step 4: compute distances to ideal solutions

$$D_i^+ = \sqrt{\frac{1}{3m} \sum_{j=1}^m [(T_{ij}^{(w)} - T_j^+)^2 + (I_{ij}^{(w)} - I_j^+)^2 + (F_{ij}^{(w)} - F_j^+)^2]} \quad (9)$$

$$D_i^- = \sqrt{\frac{1}{3m} \sum_{j=1}^m [(T_{ij}^{(w)} - T_j^-)^2 + (I_{ij}^{(w)} - I_j^-)^2 + (F_{ij}^{(w)} - F_j^-)^2]} \quad (10)$$

Table 4 Neutrosophic normalized decision table (Static Thrust, MTOW, Fineness Ratio, Aspect Ratio)

Model	Static thrust (kN)	MTOW	Fine-ness ratio	Aspect ratio
A300-600R	(0.93, 0.1, 0.07)	(0.59, 0.3, 0.41)	(0.34, 0.2, 0.66)	(0.33, 0.2, 0.67)
A310-300	(0.83, 0.2, 0.17)	(0.65, 0.3, 0.35)	(0.16, 0.3, 0.84)	(0.67, 0.2, 0.33)
A319-100	(0.29, 0.4, 0.71)	(0.92, 0.1, 0.08)	(0.23, 0.3, 0.77)	(0.83, 0.1, 0.17)
A320-200	(0.34, 0.3, 0.66)	(0.88, 0.2, 0.12)	(0.35, 0.2, 0.65)	(0.83, 0.1, 0.17)
A340-500	(0.84, 0.1, 0.16)	(0.00, 0.5, 1.00)	(0.62, 0.2, 0.38)	(0.55, 0.2, 0.45)
A340-600	(0.90, 0.1, 0.10)	(0.00, 0.5, 1.00)	(0.71, 0.2, 0.29)	(0.55, 0.2, 0.45)
707-320C	(0.23, 0.5, 0.77)	(0.61, 0.4, 0.39)	(0.07, 0.5, 0.93)	(0.07, 0.5, 0.93)
727-200Adv	(0.18, 0.5, 0.82)	(0.81, 0.3, 0.19)	(0.03, 0.6, 0.97)	(0.03, 0.6, 0.97)
737-400	(0.29, 0.4, 0.71)	(0.93, 0.1, 0.07)	(0.08, 0.4, 0.92)	(0.78, 0.2, 0.22)
737-500	(0.22, 0.5, 0.78)	(0.96, 0.1, 0.04)	(0.08, 0.4, 0.92)	(0.78, 0.2, 0.22)
737-700	(0.25, 0.4, 0.75)	(0.90, 0.2, 0.10)	(0.08, 0.4, 0.92)	(0.83, 0.1, 0.17)
737-800	(0.32, 0.3, 0.68)	(0.86, 0.2, 0.14)	(0.08, 0.4, 0.92)	(0.83, 0.1, 0.17)
767-200ER	(0.93, 0.1, 0.07)	(0.57, 0.3, 0.43)	(0.34, 0.2, 0.66)	(0.44, 0.2, 0.56)
767-300ER	(1.00, 0.0, 0.00)	(0.55, 0.3, 0.45)	(0.50, 0.2, 0.50)	(0.44, 0.2, 0.56)
DC8-73	(0.29, 0.5, 0.71)	(0.62, 0.4, 0.38)	(1.00, 0.0, 0.00)	(0.28, 0.3, 0.72)
DC 9-10	(0.11, 0.6, 0.89)	(1.00, 0.0, 0.00)	(0.13, 0.5, 0.87)	(0.67, 0.2, 0.33)
DC 9-30	(0.14, 0.5, 0.86)	(0.89, 0.2, 0.11)	(0.28, 0.3, 0.72)	(0.72, 0.2, 0.28)
DC 9-40	(0.17, 0.5, 0.83)	(0.87, 0.2, 0.13)	(0.37, 0.3, 0.63)	(0.72, 0.2, 0.28)
DC 9-50	(0.17, 0.5, 0.83)	(0.85, 0.2, 0.15)	(0.45, 0.3, 0.55)	(0.72, 0.2, 0.28)
DC10-10	(0.61, 0.3, 0.39)	(0.53, 0.4, 0.47)	(0.24, 0.3, 0.76)	(0.05, 0.6, 0.95)
DC10-30	(0.85, 0.2, 0.15)	(0.31, 0.5, 0.69)	(0.24, 0.3, 0.76)	(0.05, 0.6, 0.95)
DOUG-MD-81	(0.24, 0.4, 0.76)	(0.92, 0.1, 0.08)	(0.60, 0.2, 0.40)	(1.00, 0.0, 0.00)
DOUG-MD-82	(0.27, 0.4, 0.73)	(0.89, 0.2, 0.11)	(0.60, 0.2, 0.40)	(1.00, 0.0, 0.00)
DOUG-MD-83	(0.28, 0.4, 0.72)	(0.86, 0.2, 0.14)	(0.60, 0.2, 0.40)	(1.00, 0.0, 0.00)
DOUG-MD-87	(0.27, 0.4, 0.73)	(0.92, 0.1, 0.08)	(0.42, 0.3, 0.58)	(1.00, 0.0, 0.00)
DOUG-MD-90-30	(0.34, 0.3, 0.66)	(0.90, 0.2, 0.10)	(0.66, 0.2, 0.34)	(1.00, 0.0, 0.00)
DOUG-MD-11	(1.00, 0.0, 0.00)	(0.25, 0.5, 0.75)	(0.38, 0.3, 0.62)	(0.42, 0.2, 0.58)
R170	(0.00, 1.0, 1.00)	(0.98, 0.1, 0.02)	(0.00, 1.0, 1.00)	(0.78, 0.2, 0.22)
R185	(0.02, 0.8, 0.98)	(0.98, 0.1, 0.02)	(0.09, 0.5, 0.91)	(0.78, 0.2, 0.22)
R1100	(0.02, 0.8, 0.98)	(0.97, 0.1, 0.03)	(0.18, 0.4, 0.82)	(0.78, 0.2, 0.22)
R1115	(0.02, 0.8, 0.98)	(0.96, 0.1, 0.04)	(0.18, 0.4, 0.82)	(0.78, 0.2, 0.22)
F70	(0.14, 0.5, 0.86)	(1.00, 0.0, 0.00)	(0.22, 0.3, 0.78)	(0.61, 0.2, 0.39)
F100	(0.14, 0.5, 0.86)	(0.97, 0.1, 0.03)	(0.40, 0.3, 0.60)	(0.61, 0.2, 0.39)

4.5 Step 5: compute closeness coefficient and ranking

$$CC_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (11)$$

The alternatives are ranked in descending order CC_i . A higher value indicates a better alternative.

5 Numerical analysis

We collected a transport aircraft dataset with 10 criteria, namely, Static Thrust, MTOW, Fineness Ratio, Aspect Ratio, Wing Load, Thrust/Weight Ratio, Range, and Fuel Consumption. The favorable and unfavorable criteria are mentioned in the table for selection criteria (Table 3). We also segregate the aircraft that the same companies manufacture.

We create some sets of similar aircraft models, which are made by the same company.

- Set A (AIRBUS): {A300-600R, A310-300, A319-100, A320-200, A340-500, A340-600}
- Set B (BOEING): {707-320C, 727-200Adv, 737-400, 737-500, 737-700, 737-800, 767-200ER, 767-300ER}
- Set C (Douglas): {DC8-73, DC 9-10, DC 9-30, DC 9-40, DC 9-50, DOUG-MD-81, DOUG-MD-82, DOUG-MD-83, DOUG-MD-87, DOUG-MD-90-30, DC10-10, DC10-30, DOUG-MD-11}
- Set D (BAE): {RJ70, RJ85, RJ100, RJ115}
- Set E (FOKKER): {F70, F100}

5.1 Step 1. Determination of the selection criteria table

The selection criteria table is given by Table 3.

Table 5 Neutrosophic normalized decision table (Wing Load, Thrust/Weight Ratio, Range)

Model	Wing load (kg/m ²)	Thrust/Weight ratio	Range (km)
A300-600R	(0.62, 0.3, 0.38)	(0.72, 0.1, 0.28)	(0.48, 0.2, 0.52)
A310-300	(0.65, 0.3, 0.35)	(0.75, 0.1, 0.25)	(0.52, 0.2, 0.48)
A319-100	(0.92, 0.1, 0.08)	(0.78, 0.1, 0.22)	(0.20, 0.4, 0.80)
A320-200	(0.85, 0.2, 0.15)	(0.71, 0.1, 0.29)	(0.32, 0.3, 0.68)
A340-500	(0.00, 0.5, 1.00)	(0.31, 0.3, 0.69)	(1.00, 0.0, 0.00)
A340-600	(0.00, 0.5, 1.00)	(0.44, 0.2, 0.56)	(0.88, 0.1, 0.12)
707-320C	(0.89, 0.2, 0.11)	(0.00, 0.5, 1.00)	(0.00, 1.0, 1.00)
727-200Adv	(0.77, 0.3, 0.23)	(0.01, 0.5, 0.99)	(0.24, 0.4, 0.76)
737-400	(0.70, 0.3, 0.30)	(0.79, 0.1, 0.21)	(0.32, 0.3, 0.68)
737-500	(0.93, 0.1, 0.07)	(0.81, 0.1, 0.19)	(0.14, 0.5, 0.86)
737-700	(0.95, 0.1, 0.05)	(0.30, 0.3, 0.70)	(0.45, 0.3, 0.55)
737-800	(0.87, 0.2, 0.13)	(0.45, 0.2, 0.55)	(0.40, 0.3, 0.60)
767-200ER	(0.91, 0.1, 0.09)	(0.63, 0.1, 0.37)	(0.92, 0.1, 0.08)
767-300ER	(0.89, 0.2, 0.11)	(0.70, 0.1, 0.30)	(0.80, 0.2, 0.20)
DC8-73	(0.93, 0.1, 0.07)	(0.18, 0.4, 0.82)	(0.00, 1.0, 1.00)
DC 9-10	(1.00, 0.0, 0.00)	(0.76, 0.1, 0.24)	(0.10, 0.6, 0.90)
DC 9-30	(0.84, 0.2, 0.16)	(0.26, 0.3, 0.74)	(0.17, 0.5, 0.83)
DC 9-40	(0.81, 0.2, 0.19)	(0.39, 0.2, 0.61)	(0.16, 0.5, 0.84)
DC 9-50	(0.78, 0.2, 0.22)	(0.26, 0.3, 0.74)	(0.13, 0.5, 0.87)
DC10-10	(0.97, 0.1, 0.03)	(0.50, 0.2, 0.50)	(0.48, 0.3, 0.52)
DC10-30	(0.70, 0.3, 0.30)	(0.41, 0.2, 0.59)	(0.68, 0.2, 0.32)
DOUG-MD-81	(0.87, 0.2, 0.13)	(0.42, 0.2, 0.58)	(0.15, 0.5, 0.85)
DOUG-MD-82	(0.80, 0.2, 0.20)	(0.45, 0.2, 0.55)	(0.22, 0.4, 0.78)
DOUG-MD-83	(0.73, 0.3, 0.27)	(0.38, 0.2, 0.62)	(0.28, 0.4, 0.72)
DOUG-MD-87	(0.87, 0.2, 0.13)	(0.52, 0.2, 0.48)	(0.26, 0.4, 0.74)
DOUG-MD-90-30	(0.82, 0.2, 0.18)	(0.80, 0.1, 0.20)	(0.24, 0.4, 0.76)
DOUG-MD-11	(0.00, 0.5, 1.00)	(0.59, 0.1, 0.41)	(0.92, 0.1, 0.08)
R170	(0.85, 0.2, 0.15)	(0.39, 0.2, 0.61)	(0.19, 0.5, 0.81)
R185	(0.83, 0.2, 0.17)	(0.64, 0.1, 0.36)	(0.16, 0.5, 0.84)
R1100	(0.80, 0.2, 0.20)	(0.52, 0.2, 0.48)	(0.13, 0.5, 0.87)
R1115	(0.77, 0.2, 0.23)	(0.42, 0.2, 0.58)	(0.16, 0.5, 0.84)
F70	(1.00, 0.0, 0.00)	(1.00, 0.0, 0.00)	(0.07, 0.6, 0.93)
F100	(0.94, 0.1, 0.06)	(0.56, 0.2, 0.44)	(0.09, 0.6, 0.91)

Table 6 Neutrosophic normalized decision table (Take-off, Speed, and Fuel)

Model	Take-off Dist. (m)	Speed (kg)	Fuel Cons. (kg/h)
A300-600R	(0.54, 0.3, 0.46)	(0.69, 0.1, 0.31)	(0.57, 0.3, 0.43)
A310-300	(0.67, 0.2, 0.33)	(0.70, 0.1, 0.30)	(0.62, 0.3, 0.38)
A319-100	(0.76, 0.2, 0.24)	(0.71, 0.1, 0.29)	(0.81, 0.2, 0.19)
A320-200	(0.63, 0.3, 0.37)	(0.71, 0.1, 0.29)	(0.80, 0.2, 0.20)
A340-500	(0.50, 0.3, 0.50)	(0.00, 1.0, 1.00)	(0.00, 1.0, 1.00)
A340-600	(0.50, 0.3, 0.50)	(0.00, 1.0, 1.00)	(0.00, 1.0, 1.00)
707-320C	(0.00, 1.0, 1.00)	(0.75, 0.2, 0.25)	(0.00, 1.0, 1.00)
727-200Adv	(0.47, 0.4, 0.53)	(0.78, 0.2, 0.22)	(0.67, 0.3, 0.33)
737-400	(0.65, 0.3, 0.35)	(0.70, 0.2, 0.30)	(0.76, 0.2, 0.24)
737-500	(0.80, 0.2, 0.20)	(0.70, 0.2, 0.30)	(0.72, 0.3, 0.28)
737-700	(0.76, 0.2, 0.24)	(0.00, 1.0, 1.00)	(0.00, 1.0, 1.00)
737-800	(0.68, 0.2, 0.32)	(0.00, 1.0, 1.00)	(0.00, 1.0, 1.00)
767-200ER	(0.56, 0.3, 0.44)	(0.69, 0.1, 0.31)	(0.64, 0.3, 0.36)
767-300ER	(0.31, 0.4, 0.69)	(0.68, 0.1, 0.32)	(0.62, 0.3, 0.38)
DC8-73	(0.57, 0.4, 0.43)	(0.67, 0.2, 0.33)	(0.65, 0.3, 0.35)
DC 9-10	(1.00, 0.0, 0.00)	(0.73, 0.2, 0.27)	(0.00, 1.0, 1.00)
DC 9-30	(0.80, 0.2, 0.20)	(0.74, 0.2, 0.26)	(0.81, 0.2, 0.19)
DC 9-40	(0.70, 0.2, 0.30)	(0.73, 0.2, 0.27)	(0.82, 0.2, 0.18)
DC 9-50	(0.50, 0.4, 0.50)	(0.73, 0.2, 0.27)	(0.75, 0.2, 0.25)
DC10-10	(0.40, 0.4, 0.60)	(0.62, 0.3, 0.38)	(0.31, 0.5, 0.69)
DC10-30	(0.60, 0.3, 0.40)	(0.78, 0.2, 0.22)	(0.29, 0.5, 0.71)
DOUG-MD-81	(0.64, 0.3, 0.36)	(0.72, 0.2, 0.28)	(0.83, 0.2, 0.17)
DOUG-MD-82	(0.65, 0.3, 0.35)	(0.72, 0.2, 0.28)	(0.83, 0.2, 0.17)
DOUG-MD-83	(0.53, 0.4, 0.47)	(0.72, 0.2, 0.28)	(0.84, 0.2, 0.16)
DOUG-MD-87	(0.78, 0.2, 0.22)	(0.72, 0.2, 0.28)	(0.86, 0.1, 0.14)
DOUG-MD-90-30	(1.00, 0.0, 0.00)	(0.00, 1.0, 1.00)	(0.00, 1.0, 1.00)
DOUG-MD-11	(0.57, 0.3, 0.43)	(1.00, 0.0, 0.00)	(0.37, 0.4, 0.63)
R170	(0.88, 0.1, 0.12)	(0.00, 1.0, 1.00)	(1.00, 0.0, 0.00)
R185	(0.81, 0.2, 0.19)	(0.00, 1.0, 1.00)	(0.97, 0.1, 0.03)
R1100	(0.77, 0.2, 0.23)	(0.00, 1.0, 1.00)	(0.95, 0.1, 0.05)
R1115	(0.77, 0.2, 0.23)	(0.00, 1.0, 1.00)	(0.93, 0.1, 0.07)
F70	(0.92, 0.1, 0.08)	(0.65, 0.2, 0.35)	(0.99, 0.1, 0.01)
F100	(0.70, 0.2, 0.30)	(0.64, 0.2, 0.36)	(0.96, 0.1, 0.04)

5.2 Step 2. Determination of neutrosophic normalized Decision table

Table 7 Neutrosophic Correlation Matrix for Static thrust, MTOW, Fineness ratio, Aspect ratio, Wing load, and Thrust/weight

Metric	Static Thrust	MTOW	Fineness Ratio	Aspect Ratio	Wing Load	Thrust/ Weight
Static Thrust	1.000	0.427	0.682	0.315	0.123	0.876
MTOW	0.427	1.000	0.215	0.178	0.845	0.312
Fineness Ratio	0.682	0.215	1.000	0.467	0.302	0.589
Aspect Ratio	0.315	0.178	0.467	1.000	0.154	0.223
Wing Load	0.123	0.845	0.302	0.154	1.000	0.108
Thrust/Weight	0.876	0.312	0.589	0.223	0.108	1.000
Range	0.543	0.387	0.421	0.298	0.465	0.502
Take-off Dist	0.231	0.678	0.187	0.112	0.723	0.198
Speed	0.654	0.289	0.512	0.345	0.287	0.601
Fuel Cons	0.412	0.765	0.324	0.201	0.812	0.387

Table 8 Neutrosophic Correlation Matrix for Range, Take-off, Speed, and Fuel consumption

Metric	Range	Take-off	Speed	Fuel Cons.
Static Thrust	0.543	0.231	0.654	0.412
MTOW	0.387	0.678	0.289	0.765
Fine-ness Ratio	0.421	0.187	0.512	0.324
Aspect Ratio	0.298	0.112	0.345	0.201
Wing Load	0.465	0.723	0.287	0.812
Thrust/Weight	0.502	0.198	0.601	0.387
Range	1.000	0.456	0.587	0.498
Take-off Dist	0.456	1.000	0.321	0.734
Speed	0.587	0.321	1.000	0.412
Fuel Cons	0.498	0.734	0.412	1.000

Table 9 Measures of conflicts and weights of criteria

Criterion	C_j	w_j
Static Thrust (kN)	1.4278	0.1428
MTOW	1.3562	0.1356
Fine-ness Ratio	1.5893	0.1589
Aspect Ratio	1.2124	0.1212
Wing Load (kg/m ²)	1.4786	0.1479
Thrust/Weight Ratio	1.6321	0.1632
Range (km)	1.3025	0.1303
Take-off Dist. (m)	1.2457	0.1246
Speed (kt)	1.3874	0.1387
Fuel Cons. (kg/h)	1.5239	0.1524

The neutrosophic normalized decision table for alternatives in terms of crisp values is given by Tables 4, 5, and 6.

5.3 Step 3. Determination of pairwise correlation coefficient tables

The pairwise correlation coefficient determines the relationship between any two criteria. The pairwise correlation coefficient between each pair of criteria for all alternatives is given by Tables 7 and 8.

The correlation coefficient of two neutrosophic sets is calculated by formula

$$\rho(\mathcal{A}, \mathcal{B}) = \frac{C_{\mathcal{N}}(\mathcal{A}, \mathcal{B})}{\sqrt{V(\mathcal{A}) \cdot V(\mathcal{B})}} \quad (12)$$

Where

$$C_{\mathcal{N}}(\mathcal{A}, \mathcal{B}) = \sum_{i=1}^n [T_{\mathcal{A}}(x_i) \cdot T_{\mathcal{B}}(x_i) + I_{\mathcal{A}}(x_i) \cdot I_{\mathcal{B}}(x_i) + F_{\mathcal{A}}(x_i) \cdot F_{\mathcal{B}}(x_i)] \quad (13)$$

and

$$V(\mathcal{A}) = \sum_{i=1}^n [T_{\mathcal{A}}^2(x_i) + I_{\mathcal{A}}^2(x_i) + F_{\mathcal{A}}^2(x_i)] \quad (14)$$

and

$$V(\mathcal{B}) = \sum_{i=1}^n [T_{\mathcal{B}}^2(x_i) + I_{\mathcal{B}}^2(x_i) + F_{\mathcal{B}}^2(x_i)] \quad (15)$$

5.4 Step 4. Determination of measures of conflicts and weights of criteria

The measure of conflicts and index values are calculated by using the Eq. 6 and weights of criteria are calculated by using the Eq. 7. Measures of conflicts and weights of criteria are summarized by Table 9.

$$\text{Where: } C_j = \sigma_j \sum_{j=1}^n (1 - C_{ij}) \quad \text{for } i = 1, 2, \dots, n$$

$$w_j = \frac{C_j}{\sum_{j=1}^n C_j} \quad \text{for } j = 1, 2, \dots, n$$

σ_j = Standard deviation of criterion j in the correlation matrix

5.5 Step 5. Determination of Neutrosophic normalized weighted decision table

To obtain the weighted decision table, the weights of each criterion will be multiplied with corresponding neutrosophic set of Tables 4, 5, and 6. The weighted decision table is given by Tables 10, 11, and 12. F and UF denote the favorable and unfavorable criterion.

$$\bar{n}_{ij}^{w_j} = \langle 1 - (1 - T_{\bar{n}_{ij}})^{w_j}, I_{\bar{n}_{ij}}^{w_j}, F_{\bar{n}_{ij}}^{w_j} \rangle, \quad w_j > 0, \quad 1 \leq i \leq n, \quad 1 \leq j \leq m$$

Where:

- $T_{\bar{n}_{ij}}$ = Truth value from original neutrosophic table
- $I_{\bar{n}_{ij}}$ = Indeterminacy value from original table
- $F_{\bar{n}_{ij}}$ = Falsity value from original table
- w_j = Criterion weight from previous calculation

5.6 Step 6. Determination of neutrosophic relative positive ideal solution and neutrosophic negative ideal solution

The neutrosophic relative positive ideal solution (NRPIS) and neutrosophic negative ideal solution (NRNIS) are calculated on the basis of type of criteria i.e. favorable type or unfavorable type by using Step 3 of methodology section and given by Table 13.

Table 10 Weighted Neutrosophic Decision Matrix for All Models (Criteria 1–4)

Model	Static thrust (kN) (F)	MTOW (UF)	Fine-ness ratio (F)	Aspect ratio (F)
A300-600R	(0.99, 0.03, 0.01)	(0.28, 0.41, 0.72)	(0.89, 0.06, 0.11)	(0.85, 0.06, 0.15)
A310-300	(0.96, 0.06, 0.04)	(0.33, 0.41, 0.68)	(0.73, 0.10, 0.27)	(0.94, 0.06, 0.06)
A319-100	(0.81, 0.13, 0.19)	(0.46, 0.03, 0.54)	(0.78, 0.10, 0.22)	(0.96, 0.03, 0.04)
A320-200	(0.85, 0.10, 0.15)	(0.44, 0.06, 0.56)	(0.89, 0.06, 0.11)	(0.96, 0.03, 0.04)
A340-500	(0.96, 0.03, 0.04)	(0.00, 0.50, 1.00)	(0.95, 0.06, 0.05)	(0.92, 0.06, 0.08)
A340-600	(0.97, 0.03, 0.03)	(0.00, 0.50, 1.00)	(0.98, 0.06, 0.02)	(0.92, 0.06, 0.08)
707-320C	(0.78, 0.16, 0.22)	(0.31, 0.41, 0.70)	(0.73, 0.16, 0.27)	(0.73, 0.16, 0.27)
727-200Adv	(0.73, 0.16, 0.27)	(0.41, 0.41, 0.60)	(0.81, 0.13, 0.19)	(0.81, 0.13, 0.19)
737-400	(0.81, 0.13, 0.19)	(0.46, 0.03, 0.54)	(0.73, 0.13, 0.27)	(0.94, 0.06, 0.06)
737-500	(0.78, 0.16, 0.22)	(0.48, 0.03, 0.52)	(0.73, 0.13, 0.27)	(0.94, 0.06, 0.06)
737-700	(0.80, 0.13, 0.20)	(0.45, 0.06, 0.55)	(0.73, 0.13, 0.27)	(0.96, 0.03, 0.04)
737-800	(0.85, 0.10, 0.15)	(0.43, 0.06, 0.57)	(0.73, 0.13, 0.27)	(0.96, 0.03, 0.04)
767-200ER	(0.99, 0.03, 0.01)	(0.29, 0.41, 0.72)	(0.89, 0.06, 0.11)	(0.89, 0.06, 0.11)
767-300ER	(1.00, 0.00, 0.00)	(0.28, 0.41, 0.73)	(0.94, 0.06, 0.07)	(0.92, 0.06, 0.08)
DC8-73	(0.81, 0.16, 0.19)	(0.31, 0.41, 0.69)	(1.00, 0.00, 0.00)	(0.85, 0.10, 0.15)
DC 9-10	(0.73, 0.19, 0.27)	(1.00, 0.00, 0.00)	(0.78, 0.16, 0.22)	(0.94, 0.06, 0.06)
DC 9-30	(0.76, 0.16, 0.24)	(0.45, 0.06, 0.56)	(0.85, 0.10, 0.15)	(0.96, 0.03, 0.04)
DC 9-40	(0.78, 0.16, 0.22)	(0.44, 0.06, 0.57)	(0.89, 0.10, 0.11)	(0.96, 0.03, 0.04)
DC 9-50	(0.78, 0.16, 0.22)	(0.43, 0.06, 0.58)	(0.94, 0.10, 0.06)	(0.96, 0.03, 0.04)
DC10-10	(0.92, 0.10, 0.08)	(0.27, 0.41, 0.74)	(0.78, 0.10, 0.22)	(0.73, 0.19, 0.27)
DC10-30	(0.96, 0.06, 0.04)	(0.16, 0.50, 0.85)	(0.78, 0.10, 0.22)	(0.73, 0.19, 0.27)
DOUG-MD-81	(0.78, 0.13, 0.22)	(0.46, 0.03, 0.54)	(0.95, 0.06, 0.05)	(1.00, 0.00, 0.00)
DOUG-MD-82	(0.80, 0.13, 0.20)	(0.45, 0.06, 0.56)	(0.95, 0.06, 0.05)	(1.00, 0.00, 0.00)
DOUG-MD-83	(0.81, 0.13, 0.19)	(0.43, 0.06, 0.57)	(0.95, 0.06, 0.05)	(1.00, 0.00, 0.00)
DOUG-MD-87	(0.80, 0.13, 0.20)	(0.46, 0.03, 0.54)	(0.89, 0.10, 0.11)	(1.00, 0.00, 0.00)
DOUG-MD-90-30	(0.85, 0.10, 0.15)	(0.45, 0.06, 0.56)	(0.98, 0.06, 0.02)	(1.00, 0.00, 0.00)
DOUG-MD-11	(1.00, 0.00, 0.00)	(0.13, 0.50, 0.88)	(0.89, 0.10, 0.11)	(0.89, 0.06, 0.11)
R170	(0.00, 1.00, 1.00)	(0.49, 0.03, 0.51)	(0.00, 1.00, 1.00)	(0.94, 0.06, 0.06)
R185	(0.73, 0.30, 0.27)	(0.49, 0.03, 0.51)	(0.73, 0.16, 0.27)	(0.94, 0.06, 0.06)
R1100	(0.73, 0.30, 0.27)	(0.49, 0.03, 0.52)	(0.78, 0.13, 0.22)	(0.94, 0.06, 0.06)
R1115	(0.73, 0.30, 0.27)	(0.48, 0.03, 0.52)	(0.78, 0.13, 0.22)	(0.94, 0.06, 0.06)
F70	(0.76, 0.16, 0.24)	(1.00, 0.00, 0.00)	(0.78, 0.10, 0.22)	(0.92, 0.06, 0.08)
F100	(0.76, 0.16, 0.24)	(0.49, 0.03, 0.52)	(0.89, 0.10, 0.11)	(0.92, 0.06, 0.08)

5.7 Step 7. Determination of neutrosophic normalized euclidean distance and relative closeness coefficient

Measures of neutrosophic normalized euclidean distance as described in Eq. 9 are used to determine the euclidean distance of each alternative form (Table 14) the NRPIS and the NRNIS and finally relative closeness coefficient (11) and shown in Table 15.

Where:

- D_i^+ = Distance to positive ideal solution
- D_i^- = Distance to negative ideal solution
- CC_i = Closeness coefficient (higher values indicate better performance)
- m = Number of criteria (10 in this case)

6 Discussion, managerial implications, and Conclusions

According to Table 15 and Fig. 2, the DC 9-10 is observed to have the highest performance among the others. This section describes the cluster-wise (Set A to E) analysis, described by figure 3 of aircraft performance and selection of the best aircraft.

6.1 Airbus

A319-100 and A320-200 emerge as clear front-runners, which are renowned as narrow-body transport aircraft. Their closeness coefficients are 0.5234 for the A319-100 and 0.5064 for the A320-200. On the other hand, the A340-500 and A340-600 models are positioned at the lower end of the spectrum, with CC values of 0.3856 and 0.4002,

Table 11 Weighted Neutrosophic Decision Matrix for All Models (Criteria 5-7)

Model	Wing load (kg/m ²) (UF)	Thrust/Weight (F)	Range (km) (F)
A300-600R	(0.31, 0.41, 0.69)	(0.98, 0.03, 0.02)	(0.92, 0.06, 0.08)
A310-300	(0.33, 0.41, 0.68)	(0.99, 0.03, 0.01)	(0.94, 0.06, 0.07)
A319-100	(0.46, 0.03, 0.54)	(0.99, 0.03, 0.01)	(0.73, 0.13, 0.27)
A320-200	(0.43, 0.06, 0.58)	(0.98, 0.03, 0.02)	(0.85, 0.10, 0.15)
A340-500	(0.00, 0.50, 1.00)	(0.81, 0.10, 0.19)	(1.00, 0.00, 0.00)
A340-600	(0.00, 0.50, 1.00)	(0.89, 0.06, 0.11)	(0.96, 0.03, 0.04)
707-320C	(0.45, 0.41, 0.56)	(0.00, 0.50, 1.00)	(0.00, 1.00, 1.00)
727-200Adv	(0.39, 0.41, 0.62)	(0.73, 0.16, 0.27)	(0.78, 0.13, 0.22)
737-400	(0.35, 0.41, 0.65)	(0.99, 0.03, 0.01)	(0.85, 0.10, 0.15)
737-500	(0.47, 0.03, 0.53)	(0.99, 0.03, 0.01)	(0.73, 0.16, 0.27)
737-700	(0.47, 0.03, 0.54)	(0.81, 0.10, 0.19)	(0.89, 0.10, 0.11)
737-800	(0.44, 0.06, 0.57)	(0.89, 0.06, 0.11)	(0.85, 0.10, 0.15)
767-200ER	(0.45, 0.06, 0.56)	(0.95, 0.06, 0.05)	(0.98, 0.03, 0.02)
767-300ER	(0.45, 0.06, 0.56)	(0.98, 0.03, 0.02)	(0.96, 0.03, 0.04)
DC8-73	(0.47, 0.03, 0.54)	(0.73, 0.16, 0.27)	(0.00, 1.00, 1.00)
DC 9-10	(1.00, 0.00, 0.00)	(0.99, 0.03, 0.01)	(0.73, 0.19, 0.27)
DC 9-30	(0.53, 0.06, 0.48)	(0.85, 0.10, 0.15)	(0.80, 0.13, 0.20)
DC 9-40	(0.56, 0.06, 0.45)	(0.89, 0.06, 0.11)	(0.80, 0.13, 0.20)
DC 9-50	(0.59, 0.06, 0.42)	(0.85, 0.10, 0.15)	(0.73, 0.16, 0.27)
DC10-10	(0.97, 0.03, 0.03)	(0.94, 0.06, 0.07)	(0.92, 0.10, 0.08)
DC10-30	(0.72, 0.10, 0.29)	(0.89, 0.06, 0.11)	(0.95, 0.06, 0.05)
DOUG-MD-81	(0.44, 0.06, 0.57)	(0.89, 0.06, 0.11)	(0.76, 0.16, 0.24)
DOUG-MD-82	(0.47, 0.06, 0.54)	(0.91, 0.06, 0.09)	(0.85, 0.10, 0.15)
DOUG-MD-83	(0.50, 0.06, 0.50)	(0.87, 0.10, 0.13)	(0.89, 0.10, 0.11)
DOUG-MD-87	(0.44, 0.06, 0.57)	(0.95, 0.06, 0.05)	(0.89, 0.10, 0.11)
DOUG-MD-90-30	(0.47, 0.06, 0.54)	(0.99, 0.03, 0.01)	(0.85, 0.10, 0.15)
DOUG-MD-11	(0.00, 0.50, 1.00)	(0.95, 0.06, 0.05)	(0.98, 0.03, 0.02)
R170	(0.53, 0.06, 0.48)	(0.89, 0.06, 0.11)	(0.78, 0.16, 0.22)
R185	(0.55, 0.06, 0.46)	(0.95, 0.06, 0.05)	(0.73, 0.16, 0.27)
R1100	(0.57, 0.06, 0.43)	(0.94, 0.06, 0.07)	(0.73, 0.13, 0.27)
R1115	(0.60, 0.06, 0.41)	(0.91, 0.06, 0.09)	(0.80, 0.13, 0.20)
F70	(1.00, 0.00, 0.00)	(1.00, 0.00, 0.00)	(0.73, 0.19, 0.27)
F100	(0.94, 0.03, 0.06)	(0.94, 0.06, 0.07)	(0.76, 0.19, 0.24)

respectively. These results indicate that the performance of narrow-body and lower MTOW aircraft is relatively higher than that of others based on the parameters chosen in this study.

6.2 Boeing

Similar to Airbus, Boeing's narrow-body 737 aircraft places the highest priority among its other models. According to

Table 12 Weighted Neutrosophic Decision Matrix for All Models (Criteria 8-10)

Model	Take-off Dist. (m) (UF)	Speed (kt) (F)	Fuel Cons. (kg/h) (UF)
A300-600R	(0.29, 0.41, 0.71)	(0.98, 0.03, 0.03)	(0.29, 0.41, 0.71)
A310-300	(0.36, 0.41, 0.65)	(0.98, 0.03, 0.02)	(0.31, 0.41, 0.69)
A319-100	(0.38, 0.41, 0.62)	(0.98, 0.03, 0.02)	(0.41, 0.41, 0.59)
A320-200	(0.32, 0.41, 0.69)	(0.98, 0.03, 0.02)	(0.40, 0.41, 0.60)
A340-500	(0.25, 0.41, 0.75)	(0.00, 1.00, 1.00)	(0.00, 1.00, 1.00)
A340-600	(0.25, 0.41, 0.75)	(0.00, 1.00, 1.00)	(0.00, 1.00, 1.00)
707-320C	(0.00, 1.00, 1.00)	(0.99, 0.03, 0.01)	(0.00, 1.00, 1.00)
727-200Adv	(0.24, 0.41, 0.77)	(0.99, 0.03, 0.01)	(0.34, 0.41, 0.66)
737-400	(0.33, 0.41, 0.67)	(0.98, 0.03, 0.02)	(0.38, 0.41, 0.62)
737-500	(0.40, 0.41, 0.60)	(0.98, 0.03, 0.02)	(0.36, 0.41, 0.64)
737-700	(0.38, 0.41, 0.62)	(0.00, 1.00, 1.00)	(0.00, 1.00, 1.00)
737-800	(0.34, 0.41, 0.66)	(0.00, 1.00, 1.00)	(0.00, 1.00, 1.00)
767-200ER	(0.28, 0.41, 0.72)	(0.98, 0.03, 0.03)	(0.32, 0.41, 0.68)
767-300ER	(0.16, 0.41, 0.85)	(0.98, 0.03, 0.03)	(0.31, 0.41, 0.69)
DC8-73	(0.29, 0.41, 0.71)	(0.96, 0.10, 0.04)	(0.33, 0.41, 0.68)
DC 9-10	(1.00, 0.00, 0.00)	(0.99, 0.03, 0.01)	(0.00, 1.00, 1.00)
DC 9-30	(0.40, 0.41, 0.60)	(0.99, 0.03, 0.01)	(0.41, 0.41, 0.59)
DC 9-40	(0.35, 0.41, 0.65)	(0.99, 0.03, 0.01)	(0.40, 0.41, 0.60)
DC 9-50	(0.25, 0.41, 0.75)	(0.99, 0.03, 0.01)	(0.45, 0.41, 0.55)
DC10-10	(0.20, 0.41, 0.80)	(0.96, 0.10, 0.04)	(0.16, 0.50, 0.85)
DC10-30	(0.30, 0.41, 0.70)	(0.99, 0.03, 0.01)	(0.15, 0.50, 0.85)
DOUG-MD-81	(0.33, 0.41, 0.67)	(0.99, 0.03, 0.01)	(0.41, 0.41, 0.59)
DOUG-MD-82	(0.33, 0.41, 0.67)	(0.99, 0.03, 0.01)	(0.41, 0.41, 0.59)
DOUG-MD-83	(0.27, 0.41, 0.73)	(0.99, 0.03, 0.01)	(0.40, 0.41, 0.60)
DOUG-MD-87	(0.39, 0.41, 0.61)	(0.99, 0.03, 0.01)	(0.39, 0.41, 0.61)
DOUG-MD-90-30	(1.00, 0.00, 0.00)	(0.00, 1.00, 1.00)	(0.00, 1.00, 1.00)
DOUG-MD-11	(0.29, 0.41, 0.71)	(1.00, 0.00, 0.00)	(0.19, 0.41, 0.81)
R170	(0.44, 0.03, 0.56)	(0.00, 1.00, 1.00)	(1.00, 0.00, 0.00)
R185	(0.40, 0.41, 0.60)	(0.00, 1.00, 1.00)	(0.97, 0.03, 0.03)
R1100	(0.39, 0.41, 0.61)	(0.00, 1.00, 1.00)	(0.95, 0.03, 0.05)
R1115	(0.39, 0.41, 0.61)	(0.00, 1.00, 1.00)	(0.93, 0.03, 0.07)
F70	(0.46, 0.03, 0.54)	(0.96, 0.10, 0.04)	(0.99, 0.03, 0.01)
F100	(0.35, 0.41, 0.65)	(0.95, 0.10, 0.05)	(0.96, 0.03, 0.04)

the CC values, the 737-500 (0.5203) and 737-400 (0.5052) benefit from efficient fuel consumption, compact runway footprints, and robust thrust profiles. The 737-700 and 737-800, mid-tier models, receive CC values of around 0.420–0.480, while the longer-range 767 variants receive 0.4801 for the 767-200ER and 0.4652 for the 767-300ER. The 767 variant remains respectable, but loses ground due to higher wing loadings and thrust demands. However, due to significantly higher fuel consumption, the 727-200Adv (0.3701) and 707-320C (0.3284) fall to the bottom.

Table 13 Neutrosophic Relative Positive Ideal Solution (NRPIS)

Criteria	Type	NRPIS ($\mathcal{A}^+$)
Static thrust (kN)	Favorable	(1.00, 0.00, 0.00)
MTOW	Unfavorable	(0.00, 0.50, 1.00)
Fine-ness ratio	Favorable	(1.00, 0.00, 0.00)
Aspect ratio	Favorable	(1.00, 0.00, 0.00)
Wing load (kg/m ²)	Unfavorable	(0.00, 0.50, 1.00)
Thrust/Weight ratio	Favorable	(1.00, 0.00, 0.00)
Range (km)	Favorable	(1.00, 0.00, 0.00)
Take-off distance (m)	Unfavorable	(0.16, 0.50, 0.85)
Speed (kt)	Favorable	(1.00, 0.00, 0.00)
Fuel consumption (kg/h)	Unfavorable	(0.00, 0.50, 1.00)

Table 14 Neutrosophic Relative Negative Ideal Solution (NRNIS)

Criteria	Type	NRNIS ($\mathcal{A}^-$)
Static thrust (kN)	Favorable	(0.00, 1.00, 1.00)
MTOW	Unfavorable	(1.00, 0.03, 0.51)
Fine-ness ratio	Favorable	(0.00, 1.00, 1.00)
Aspect ratio	Favorable	(0.00, 1.00, 1.00)
Wing load (kg/m ²)	Unfavorable	(1.00, 0.03, 0.42)
Thrust/Weight ratio	Favorable	(0.00, 1.00, 1.00)
Range (km)	Favorable	(0.00, 1.00, 1.00)
Take-off distance (m)	Unfavorable	(1.00, 0.03, 0.54)
Speed (kt)	Favorable	(0.00, 1.00, 1.00)
Fuel consumption (kg/h)	Unfavorable	(1.00, 0.03, 0.55)

6.3 Douglas

Among the Douglas and McDonnell Douglas families, the DC-9-10 emerges as the highest-performing alternative in this study, achieving the highest closeness coefficient (CC = 0.5387). This superior performance is attributed to its operational efficiency over the short-to-medium ranges category and its comparatively low runway length requirements. Following are the MD-87 (CCi = 0.5348) and MD-83 (CCi = 0.5231), both of which exhibit optimized wing loading, fuel economy, and manageable take-off distances—parameters that align well with the neutrosophic TOPSIS model's decision structure.

6.4 Fokker

The F70 (CC = 0.5073) and F100 (CC = 0.4924) demonstrate strong results given their operational niche. The F70 has a higher score, which is achieved due to its compact structural characteristics and reduced wing loading, enabling improved performance on metrics such as thrust-to-weight ratio and fuel consumption.

6.5 BAE

The BAE RJ series presents a more modest performance profile, clustering within the lower mid-tier of evaluated alternatives (CC varies from 0.40 to 0.47). Notably, models such as the RJ100 achieve relatively acceptable results on criteria like take-off distance and fuel usage. However, all RJ variants exhibit limitations in terms of range and thrust-to-weight ratios, factors that constrain their overall evaluation when compared to contemporary narrow-body aircraft.

6.6 Managerial implications

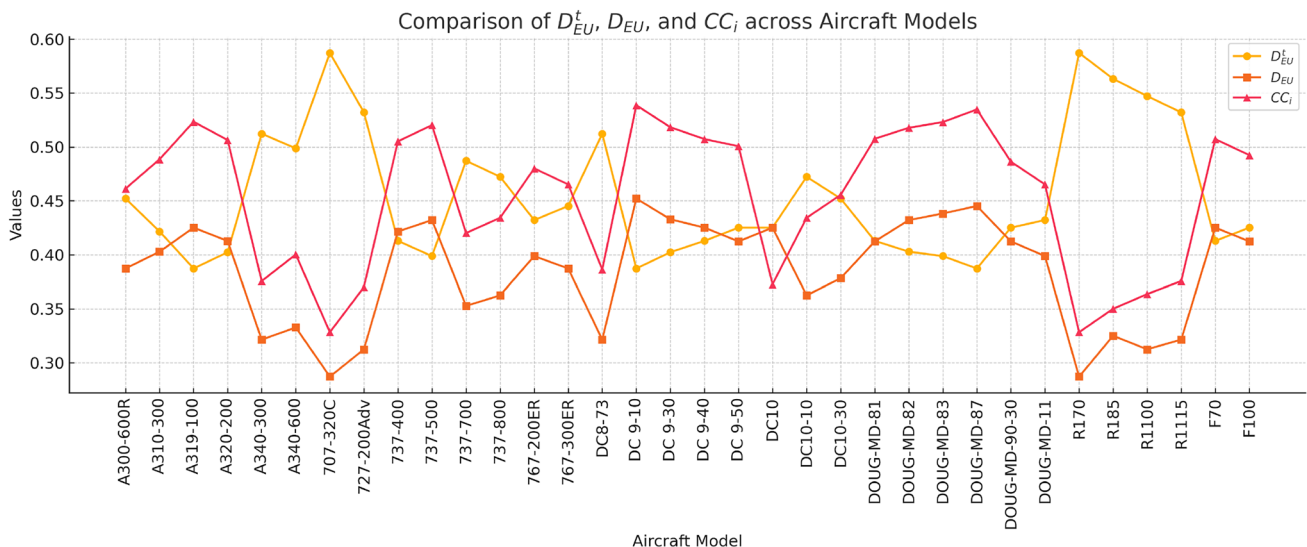
- Adoption of neutrosophic imputation can significantly improve decision quality when procurement data are incomplete or expert assessments conflict.
- Objective weights derived via neutrosophic CRITIC ensure that criteria with high discrimination power (e.g., range) and low redundancy (e.g., wing load vs. thrust/weight) receive appropriate emphasis.

Table 15 Euclidean Distances and Closeness Coefficients

Aircraft Model	D_{EU}^+	D_{EU}^-	CC_i
A300-600R	0.4521	0.3872	0.4613
A310-300	0.4215	0.4028	0.4886
A319-100	0.3872	0.4251	0.5234
A320-200	0.4023	0.4127	0.5064
A340-500	0.5123	0.3215	0.3856
A340-600	0.4987	0.3328	0.4002
707-320C	0.5872	0.2871	0.3284
727-200Adv	0.5321	0.3125	0.3701
737-400	0.4128	0.4215	0.5052
737-500	0.3987	0.4321	0.5203
737-700	0.4872	0.3528	0.4201
737-800	0.4723	0.3625	0.4342
767-200ER	0.4321	0.3987	0.4801
767-300ER	0.4452	0.3872	0.4652
DC8-73	0.5123	0.3215	0.3856
DC 9-10	0.3872	0.4521	0.5387
DC 9-30	0.4023	0.4328	0.5183
DC 9-40	0.4128	0.4251	0.5073
DC 9-50	0.4251	0.4123	0.4924
DC10-10	0.4723	0.3625	0.4342
DC10-30	0.4521	0.3782	0.4555
DOUG-MD-81	0.4123	0.4251	0.5076
DOUG-MD-82	0.4028	0.4321	0.5178
DOUG-MD-83	0.3987	0.4372	0.5231
DOUG-MD-87	0.3872	0.4452	0.5348
DOUG-MD-90-30	0.4251	0.4023	0.4863
DOUG-MD-11	0.4452	0.3872	0.4652
R170	0.5872	0.2871	0.3284
R185	0.5623	0.3028	0.3501
R1100	0.5472	0.3125	0.3637
R1115	0.5321	0.3215	0.3765
F70	0.4128	0.4251	0.5073
F100	0.4251	0.4123	0.4924

- The closeness-coefficient output provides a clear, quantitative ranking that executives can use to justify aircraft acquisitions or fleet retirements in board presentations.
- Sensitivity analysis of truth–indeterminacy–falsity thresholds allows risk-averse organizations to tune the model according to their tolerance for data uncertainty.
- Beyond aircraft selection, the framework can be adapted to any capital-intensive procurement—ships, locomotives, or industrial machinery—where performance metrics and expert judgments intersect.

Therefore, the final observation shows that the Douglas model DC 9-10 achieves the highest ranking among all other models. This study efficiently selects the best chosen aircraft under given criteria. This study is proven to be a highly efficient tool to identify the best alternatives. Under the uncertain circumstances, neutrosophic sets plays important roles for decision-making. However, some limitations of this study may be pointed out compared to dealing with a more complex and practical scenario. The neutrosophic membership functions for missing values requires highly expert judgment, which may result in subjectivity and limit reproducibility. The dataset covers 33 historical transport aircraft; extending the number to emerging unmanned or hybrid-electric models may require recalibration of criteria and imputation rules, which may change the final decision of selection. The study can be extended with a probabilistic approach (Chauhan et al. 2021) instead of fuzzy or neutrosophic sets, which may lead to a more precise decision.

**Fig. 2** Best aircraft selection among all models

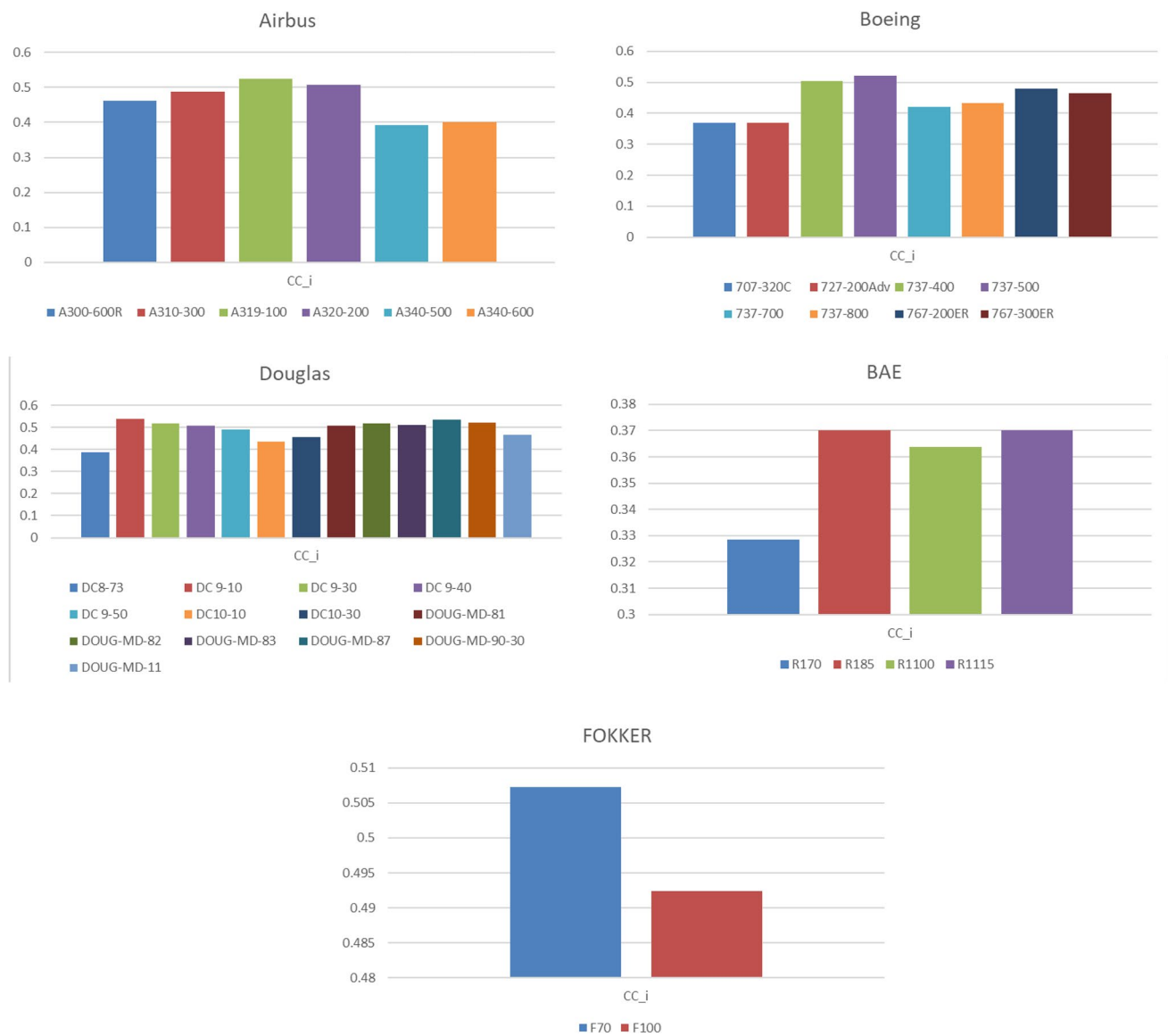


Fig. 3 Company-wise best aircraft selection

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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